

Module 25: Power BI Master Capstone Project & Career Launch Blueprint

Overview

This module serves as the culmination of the entire Power BI Master Course.

You have learned:

- Power Query
- Data Modeling
- DAX
- Visualization
- Power BI Service
- Security
- Performance Optimization
- Governance
- Fabric
- Enterprise Architecture

Now it is time to combine all these skills into a complete enterprise-grade project and build a roadmap for launching your career as a Data Analyst, BI Analyst, or Power BI Developer.

This capstone simulates a real-world consulting engagement where you must gather requirements, design architecture, build dashboards, optimize performance, secure data, document the solution, and present business recommendations.

Learning Objectives

After completing this module, you will be able to:

- Deliver end-to-end Power BI solutions
- Handle enterprise reporting requirements
- Design scalable architectures
- Apply advanced DAX techniques
- Optimize large datasets
- Implement security and governance
- Build a professional portfolio project
- Present findings to stakeholders
- Prepare for analytics careers

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Capstone Introduction

Goal

Build a complete enterprise reporting solution.

Skills Applied

```
Data Collection
  ↓
Data Cleaning
  ↓
Modeling
  ↓
DAX
  ↓
Visualization
  ↓
Optimization
  ↓
Security
```

↓
Deployment

2 Business Scenario

Company

Global Retail Corporation

Industry

Retail & E-Commerce

Challenges

Revenue Growth Slowing

Customer Retention Declining

Regional Performance Differences

Product Profitability Issues

Inventory Management Challenges

Objective

Build a comprehensive analytics solution for executives.

3 Business Requirements

Executives require answers to:

Sales

Which regions generate the most revenue?

Customers

Which customers drive profitability?

Products

Which products underperform?

Operations

Where is inventory inefficient?

Finance

What drives profit margins?

Stakeholder Analysis

Executive Team

Needs:

- High-level KPIs
- Strategic insights

Sales Managers

Need:

- Regional performance

Product Managers

Need:

- Product analytics

Finance Team

Need:

- Profitability metrics

Operations Team

Need:

- Inventory monitoring

5 KPI Identification

Executive KPIs

Revenue

Profit

Profit Margin %

Growth %

Customer Count

Average Order Value

Retention Rate

Inventory Turnover

KPI Framework

Financial

+

Customer

+

Operational

6 Data Architecture Design

Sources

Sales Data

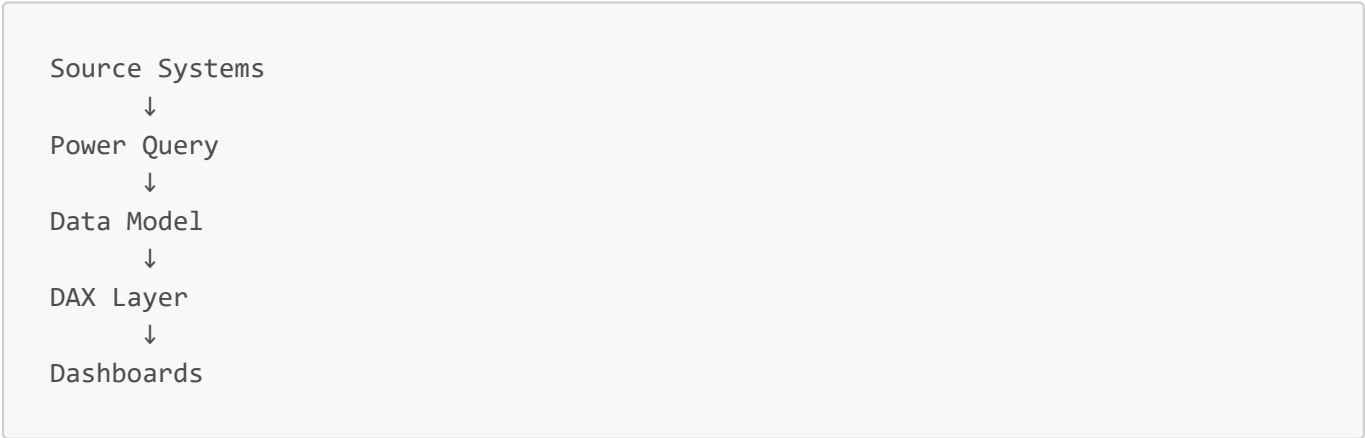
Customer Data

Product Data

Inventory Data

Regional Data

Architecture



7 Data Preparation Strategy

Tasks

Remove Duplicates

Handle Missing Values

Standardize Formats

Validate Data Quality

Create Date Table

Power Query Deliverables

Clean analytical dataset.

8 Data Modeling Design

Star Schema

```
DimCustomer
  ↓
FactSales
  ↑
DimProduct
  ↑
DimDate
  ↑
DimRegion
```

Benefits

Better Performance

Simpler DAX

Easier Maintenance

9 DAX Development Plan

Core Measures

Total Revenue

Total Profit
Profit Margin %
Customer Count

Advanced Measures

YTD Revenue
Revenue Growth %
Market Share %
Customer Ranking
Product Ranking

10

Dashboard Development

Page 1

Executive Overview

KPI Cards

- Revenue
- Profit
- Margin %
- Growth %

Trend Analysis
Monthly Revenue

Regional Performance
Map Visualization

Page 2
Sales Analysis

Product Performance

Customer Analysis

Regional Breakdown

Page 3
Inventory Analytics

Stock Levels

Inventory Turnover

Product Availability

Page 4
Profitability Dashboard

Product Profitability

Regional Profitability

1 1 Advanced Analytics
--

Decomposition Tree

Revenue Analysis

Key Influencers

Customer Retention Analysis

Forecasting

Future Revenue Trends

AI Visuals

Automated Insights

1

2

Security Implementation

Dynamic RLS

Restrict users by region.

Example

```
USERPRINCIPALNAME()
```

Roles

Executive

Full Access

Regional Manager

Regional Access

Sales Representative

Limited Access

1 3 Performance Optimization

Apply

Star Schema

Optimized DAX

Incremental Refresh

Aggregations

Performance Analyzer

Goal

Sub-5-second report responsiveness.

1 4 Deployment Strategy

Environment Structure

```
graph TD;
  Development --> Testing;
  Testing --> Production;
```

Deployment Pipeline

Controlled releases.

Validation

Security

Performance

Accuracy

1 5 Documentation Standards

Required Documents

README.md

Project Overview

Business Requirements

Data Dictionary

KPI Definitions

Technical Documentation

User Guide

1 6 Business Presentation Framework

Slide 1

Business Problem

Slide 2

Data Overview

Slide 3

KPIs

Slide 4

Dashboard Walkthrough

Slide 5

Key Insights

Slide 6

Recommendations

Slide 7

Expected Business Impact

1 7 GitHub Portfolio Packaging

Repository Structure

```
PowerBI-Capstone
├── Dataset
├── Documentation
├── SQL
├── PowerBI
├── Screenshots
├── Presentation
└── README.md
```

Include

Dashboard Screenshots

Documentation

Business Insights

Technical Details

1 8 Career Launch Blueprint

Beginner Stage

Learn:

- Excel
- SQL

- Power BI
-

Intermediate Stage

Learn:

- Advanced DAX
 - Data Modeling
 - Power BI Service
-

Advanced Stage

Learn:

- Fabric
 - Azure
 - Data Engineering
-

Goal

Become an end-to-end analytics professional.

1**9**

90-Day Job Preparation Plan

Month 1

Complete Portfolio

Build:

- Sales Dashboard
 - HR Dashboard
-

Learn SQL

Daily Practice.

Month 2

Build Additional Projects

- Finance Dashboard
 - Customer Analytics
-

Interview Preparation

Power BI Questions.

Month 3

Portfolio Refinement

Resume Building

Job Applications

Mock Interviews

2 **0** Final Skills Checklist

Data Preparation

☒ Power Query

Data Modeling

☒ Star Schema

☒ Relationships

DAX

☒ Measures

☒ Time Intelligence

☒ Advanced DAX

Visualization

☒ Dashboard Design

Service

☒ Publishing

☒ Sharing

Security

✓ RLS

Performance

✓ Optimization

Governance

✓ Administration

Fabric

✓ OneLake

✓ Lakehouse

2 **1** Deliverables

Technical Deliverables

PBIX File

Documentation

Data Dictionary

KPI Definitions

Security Documentation

Performance Report

Portfolio Deliverables

GitHub Repository

Dashboard Screenshots

Presentation Deck

Case Study



Final Capstone Project

Enterprise Retail Analytics Platform

Requirements

Data Sources

- Sales
 - Customers
 - Products
 - Inventory
 - Regions
-

Data Modeling

- Star Schema
 - Date Dimension
 - Surrogate Keys
-

DAX

- Revenue
 - Profit
 - Margin
 - Growth
 - Ranking
-

Dashboard Pages

1. Executive Dashboard
 2. Sales Analytics
 3. Customer Analytics
 4. Inventory Analytics
 5. Profitability Analysis
-

Security

Dynamic RLS

Performance

Incremental Refresh

Deployment

Power BI Service

Documentation

Complete Project Documentation



Module Summary

Congratulations. You have completed the complete Power BI Master Course.

Throughout the course, you learned:

- ✓ Power BI Fundamentals
- ✓ Data Preparation
- ✓ Power Query
- ✓ Data Modeling
- ✓ DAX
- ✓ Advanced DAX
- ✓ Visualization
- ✓ Dashboard Design
- ✓ Power BI Service
- ✓ Security & RLS
- ✓ Performance Optimization
- ✓ Governance
- ✓ DevOps
- ✓ Dataflows
- ✓ Advanced Modeling

- ✓ Portfolio Development
- ✓ Interview Preparation
- ✓ Microsoft Fabric
- ✓ Enterprise Analytics Architecture

You now possess the knowledge required to build professional Power BI solutions from raw data to executive dashboards.



Recommended Next Steps

Build 5–10 Portfolio Projects

Earn PL-300 Certification

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Learn Advanced SQL

Learn Microsoft Fabric

Learn Azure Fundamentals

Contribute to GitHub Portfolio

:contentReference[oaicite:1]{index=1}

Prepare for Data Analyst Interviews

Apply Skills in Real Business Projects

🏁 End of Course

Power BI Mastery Roadmap: Beginner → Advanced → Enterprise → Career Ready